

Multiphysics (from Akhilesh's Notes)

Coupled Mechano-Thermo-Chemical Problem $\Pi(F, T, C)$

① <u>Mechanics</u>: $\nabla \cdot P(F, S, T) = 0$ $\uparrow$ first Piola-Kirchhoff stress tensor $\downarrow$ deformation gradient	Coupling
② <u>Thermal</u>: $\frac{\partial T}{\partial t} = -\nabla \cdot J_T = D_T \nabla^2 \mu_T$ $\nwarrow$ conservation $\nearrow$ Fourier's law $\searrow$ Thermal Potential	$P = \frac{\partial \Pi}{\partial F}$ $\mu_T = \frac{\partial \Pi}{\partial T}$
③ <u>Species Transport</u>: $\frac{\partial C}{\partial t} = -\nabla \cdot J_C = D_C \nabla^2 \mu_C$ $\nwarrow$ concentration of chemical species $\nearrow$ mass diffusion coeff. $\searrow$ Chemical Potential	$\mu_C = \frac{\partial \Pi}{\partial C}$

Deriv of strong form PDEs -

The coupled free energy density Π :

$$\Pi(C, T, u) = \frac{1}{2} C^2 + \frac{1}{2} T^2 + \frac{1}{2} K_{ijkl} \epsilon_{ij}^{el} \epsilon_{kl}^{el}$$

where, ϵ^{el} = elastic strain = total - (chemical eigenstrain + thermal eigenstrain)

$$\epsilon^{el} = \epsilon - \epsilon^{ch} - \epsilon^{th}$$

$$\epsilon^{ch} = \alpha_c \cdot C$$

$$\epsilon^{th} = \alpha_T \cdot T$$

$$\therefore \epsilon_{ij}^{el} = \epsilon_{ij} - \alpha_{ij}^C C - \alpha_{ij}^T T$$

$$\epsilon_{ij} = \frac{1}{2} \left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} \right)$$

To derive each strong form PDEs, take derivative of Π wrt. the respective field

① Chemical: $\mu_c = \frac{\partial \Pi}{\partial c} = 0 \text{ in } \Omega$

$$\frac{\partial \Pi}{\partial c} = c + \frac{1}{2} \Phi_{ijkl} (-\alpha_{ij}^c) \epsilon_{kl}^{el} + \frac{1}{2} \Phi_{ijkl} \epsilon_{ij}^{el} (-\alpha_{kl}^c)$$

use symmetry:

$$\frac{\partial \Pi}{\partial c} = c - \Phi_{ijkl} \epsilon_{ij}^{el} \alpha_{kl}^c$$

$\Rightarrow$

$$\mu_c = c - \Phi_{ijkl} \epsilon_{ij}^{el} \alpha_{kl}^c$$

plug this into $\downarrow$

Next: Conservation law:

Fick's Law

$$\frac{\partial c}{\partial t} = -\nabla \cdot J_c = \nabla \cdot (D_c \nabla \mu_c)$$

$$\textcircled{1} \left\{ \begin{array}{l} \Rightarrow \frac{\partial c}{\partial t} = D_c \nabla \cdot \nabla \mu_c \text{ on } \Omega \\ \text{Dirichlet BC}^n: \\ C = C_0 \text{ on } \partial \Omega_g \\ \text{Neumann BC}^n: \\ \nabla \mu_c \cdot n = J_c \text{ on } \partial \Omega_h \end{array} \right.$$

② Thermal: $\mu_T = \frac{\partial \Pi}{\partial T} = 0 \text{ in } \Omega$

$$\frac{\partial \Pi}{\partial T} = T + \frac{1}{2} \Phi_{ijkl} (-\alpha_{ij}^T) \epsilon_{kl}^{el} + \frac{1}{2} \Phi_{ijkl} \epsilon_{ij}^{el} (-\alpha_{kl}^T)$$

use symmetry: $\frac{\partial \Pi}{\partial T} = \mu_T = T - \Phi_{ijkl} \epsilon_{ij}^{el} \alpha_{kl}^T$

Next: Plug μ_T into conservⁿ law:

$$\frac{\partial T}{\partial t} = -\nabla \cdot J_T = \nabla \cdot (D_T \nabla \mu_T)$$

$$\textcircled{2} \left\{ \begin{array}{l} \Rightarrow \frac{\partial T}{\partial t} = D_T \nabla \cdot (\nabla \mu_T) \text{ on } \Omega \\ \text{Dirichlet BC}^n: T = T_0 \text{ on } \partial \Omega_g \\ \text{Neumann BC}^n: \\ \nabla \mu_T \cdot n = J_T \text{ on } \partial \Omega_h \end{array} \right.$$

③ Mechanics: $\sigma_{ij} = \frac{\partial \Pi}{\partial \epsilon_{ij}}$

$$\frac{\partial \Pi}{\partial \epsilon_{ij}} = \frac{1}{2} C_{ijkl} \cdot (1) \cdot \epsilon_{kl}^{el} + \frac{1}{2} C_{ijkl} \cdot \epsilon_{ij}^{el} \cdot (1)$$

use symmetry:

$$\frac{\partial \Pi}{\partial \epsilon_{ij}} = C_{ijkl} \epsilon_{kl}^{el}$$

$$\Rightarrow \sigma_{ij} = \frac{\partial \Pi}{\partial \epsilon_{ij}} = C_{ijkl} (\epsilon_{kl} - \alpha_{kl}^c c - \alpha_{kl}^T T)$$

Next: Apply Euler Lagrange (force balance):

$$\sigma_{ij,j} = 0 \quad \text{in } \Omega$$

$$[C_{ijkl} (\epsilon_{kl} - \alpha_{kl}^c c - \alpha_{kl}^T T)]_{,j} = 0$$

$$\textcircled{3} \left\{ \begin{array}{l} \Rightarrow \sigma_{ij,j} = 0 \quad \text{on } \Omega \end{array} \right. \quad \begin{array}{l} \text{BC's: Dirichlet BC}^n: \bar{u}_i = u_{0i} \text{ on } \partial\Omega_D \\ \text{Neumann BC}^n: \bar{\sigma}_{ij} n_j = h_i \text{ on } \partial\Omega_N \end{array}$$

We now have all three strong forms: ① ; ② ; ③

Let's now proceed to derive weak forms:

① Chemical:

multiply the strong form ① by w^c & integrate -

$$\int_{\Omega} w^c \frac{dc}{dt} dV = \int_{\Omega} w^c D_c (\nabla \cdot \nabla u_c) dV$$

↓ Product Rule

$$= \int_{\Omega} \nabla \cdot (w^c D_c \nabla u_c) dV - \int_{\Omega} \nabla w^c D_c \nabla u_c dV$$

↓ Divergence Thm

$$= \int_{\partial\Omega} w^c D_c \nabla u_c \cdot \hat{n} dS - \int_{\Omega} \nabla w^c D_c \nabla u_c dV$$

Neumann BC on $\partial\Omega^h$
zero on $\partial\Omega^f$

$$\Rightarrow \int_{\Omega} w^c \frac{dc}{dt} dV = \int_{\partial\Omega^h} w^c D_c J_c dS - \int_{\Omega} \nabla w^c D_c \nabla u_c dV$$

Rearrange:

$$\Rightarrow \int_{\Omega} w^c \frac{dc}{dt} dV + \int_{\Omega} \nabla w^c D_c \nabla u_c dV = \int_{\partial\Omega^h} w^c D_c J_c dS$$

Next: use u_c derived before to expose coupling: $(E_{ij}^T - \alpha_{ij}^T \Phi - \alpha_{ij}^c C)$

we have: $u_c = C - \Phi_{ijkl} E_{ij}^{el} \alpha_{kl}^c$

$$\nabla u_c = \nabla C - \Phi_{ijkl} (\nabla E_{ij} - \alpha_{ij}^c \nabla C - \alpha_{ij}^T \nabla T) \alpha_{kl}^c$$

$$\Rightarrow \int_{\Omega} w^c \frac{dc}{dt} dV + \int_{\Omega} \nabla w^c D_c \nabla C dV - \int_{\Omega} \nabla w^c D_c \Phi_{ijkl} \nabla E_{ij} \alpha_{kl}^c dV$$

$$\begin{aligned}
 \Rightarrow \int_{\Omega} \underbrace{w^c \frac{dc}{dt}}_{M_{cc}} dV &+ \int_{\Omega} \underbrace{\nabla w^c D_c \nabla c}_{K_{cc}} dV - \int_{\Omega} \underbrace{\nabla w^c D_c \Phi_{ijke} \nabla E_{ij} \alpha_{ke}^c}_{K_{cu}} dV \\
 &+ \int_{\Omega} \underbrace{\nabla w^c D_c \Phi_{ijke} \alpha_{ij}^c \nabla c \alpha_{ke}^c}_{K_{cc}} dV + \int_{\Omega} \underbrace{\nabla w^c D_c \Phi_{ijke} \alpha_{ij}^T \nabla T \alpha_{ke}^c}_{K_{cT}} dV \\
 &= \int_{\partial \Omega^h} \underbrace{w^c D_c J_c}_{f_c} dS
 \end{aligned}$$

$$\begin{aligned}
 \Rightarrow \left. \begin{aligned}
 &\int_{\Omega} \underbrace{w^c \frac{dc}{dt}}_{M_{cc}} dV + \int_{\Omega} \underbrace{\nabla w^c D_c (1 + \Phi_{ijke} \alpha_{ij}^c \alpha_{ke}^c)}_{K_{cc}} \nabla c dV \\
 &+ \int_{\Omega} \underbrace{\nabla w^c D_c \Phi_{ijke} \alpha_{ij}^T \alpha_{ke}^c \nabla T}_{K_{cT}} dV \\
 &- \int_{\Omega} \underbrace{\nabla w^c D_c \Phi_{ijke} \alpha_{ke}^c \nabla E_{ij}}_{K_{cu}} dV
 \end{aligned} \right\} = \int_{\partial \Omega^h} \underbrace{w^c D_c J_c}_{f_c} dS
 \end{aligned}$$

$\hookrightarrow$ ignore because of ∇E_{ij}

$\swarrow$ zero as there is noumann BC

② Thermal:

Multiply the strong form ② by w^T & integrate -

$$\int_{\Omega} w^T \frac{dT}{dt} dV = \int_{\Omega} w^T D_T \nabla \cdot (\nabla u_T) dV$$

Product Rule

$$= \int_{\Omega} \nabla \cdot (w^T D_T \nabla u_T) dV - \int_{\Omega} \nabla w^T D_T \nabla u_T dV$$

Divergence Thm

$$= \int_{\partial\Omega} w^T D_T \nabla u_T \hat{n} dS - \int_{\Omega} \nabla w^T D_T \nabla u_T dV$$

Neumann BC on $\partial\Omega^h$
Zero on $\partial\Omega^*$

$$\Rightarrow \int_{\Omega} w^T \frac{\partial T}{\partial t} dV = \int_{\partial\Omega^h} w^T D_T J_T dS - \int_{\Omega} \nabla w^T D_T \nabla u_T dV$$

Rearrange:

$$\Rightarrow \int_{\Omega} w^T \frac{\partial T}{\partial t} dV + \int_{\Omega} \nabla w^T D_T \nabla u_T dV = \int_{\partial\Omega^h} w^T D_T J_T dS$$

Next: use u_T derived before to expose coupling:

we have: $u_T = T - \Phi_{ijkl} E_{ij}^e \alpha_{kl}^T$

$$\nabla u_T = \nabla T - \Phi_{ijkl} (\nabla E_{ij} - \alpha_{ij}^e \nabla c - \alpha_{ij}^T \nabla T) \alpha_{kl}^T$$

$$\begin{aligned}
 \textcircled{P^T} \Rightarrow & \int_{\Omega} w^T \frac{\partial \Gamma}{\partial t} dV + \int_{\Omega} \nabla w^T D_T \nabla T dV - \int_{\Omega} \nabla w^T D_T \phi_{ijke} \nabla e_{ij} \alpha_{ke}^T dV \\
 & + \int_{\Omega} \nabla w^T D_T \phi_{ijke} \alpha_{ij}^c \nabla c \alpha_{ke}^T dV + \int_{\Omega} \nabla w^T D_T \phi_{ijke} \alpha_{ij}^T \nabla T \alpha_{ke}^T dV \\
 & = \int_{\partial \Omega^h} w^T D_T J_T ds
 \end{aligned}$$

$$\begin{aligned}
 \Rightarrow & \underbrace{\int_{\Omega} w^T \frac{\partial \Gamma}{\partial t} dV}_{M_{TT}} + \underbrace{\int_{\Omega} \nabla w^T D_T (1 + \phi_{ijke} \alpha_{ij}^T \alpha_{ke}^T) \nabla T dV}_{K_{TT}} \\
 & + \underbrace{\int_{\Omega} \nabla w^T D_T \phi_{ijke} \alpha_{ij}^c \alpha_{ke}^T \nabla c dV}_{K_{TC}} \\
 & - \underbrace{\int_{\Omega} \nabla w^T D_T \phi_{ijke} \alpha_{ke}^T \nabla e_{ij} dV}_{K_{Te}} \quad \left. \vphantom{\int_{\Omega} \nabla w^T D_T \phi_{ijke} \alpha_{ij}^c \alpha_{ke}^T \nabla c dV} \right\} = \underbrace{\int_{\partial \Omega^h} w^T D_T J_T ds}_{f_T}
 \end{aligned}$$

$\downarrow$
 zero as there is no Neumann BC

$\rightarrow$ ignore bias of ∇e_{ij}

③ Mechanics:

Multiply the strong form ③ by w_i^u & Integrate -

$$\int_{\Omega} w_i^u \sigma_{ij,j} dV = 0$$

Product Rule

$$\int_{\Omega} w_i^u \sigma_{ij,j} dV$$

$$\int_{\Omega} (w_i^u \sigma_{ij})_{,j} dV - \int_{\Omega} w_{i,j}^u \sigma_{ij} dV = 0$$

Divergence Thm

$$\int_{\partial\Omega} w_i^u \sigma_{ij} n_j ds - \int_{\Omega} w_{i,j}^u \sigma_{ij} dV = 0$$

Neumann BC on $\partial\Omega^n$
zero on $\partial\Omega^D$

$$\Rightarrow \int_{\Omega} w_{i,j}^u \sigma_{ij} dV = \int_{\partial\Omega^n} w_i^u h_i ds$$

Next- use σ_{ij} derived before to introduce coupling:

$$\Rightarrow \int_{\Omega} w_{i,j}^u \Phi_{ijke} (E_{ke} - \alpha_{ke}^c c - \alpha_{ke}^T T) dV = \int_{\partial\Omega^n} w_i^u h_i ds$$

$$\textcircled{Ru} \Rightarrow \int_{\Omega} w_{i,j}^u \Phi_{ijke} E_{ke} dV + \int_{\Omega} w_{i,j}^u \Phi_{ijke} E_{ke} dV - \int_{\Omega} w_{i,j}^u \Phi_{ijke} \alpha_{ke}^c c dV$$

K_{uu}

K_{uc}

$$- \int_{\Omega} w_{i,j}^u \Phi_{ijke} \alpha_{ke}^T T dV = \int_{\partial\Omega^n} w_i^u h_i ds$$

K_{uT}

f_u

zero as there is no Neumann BC

The full Monolithic Weak Form:

$$\hat{R} = R_c + R_T + R_u$$

where, Summarize:

$$R_c \Rightarrow M_{cc}\dot{c} + K_{cc}c + K_{cT}T + K_{cu}u = f_c$$

$$R_T \Rightarrow M_{TT}\dot{T} + K_{TT}T + K_{Tc}c + K_{Tu}u = f_T$$

$$R_u \Rightarrow K_{uu}u + K_{uc}c + K_{uT}T = f_u$$

$$\begin{bmatrix} M_{cc} & M_{TT} & 0 \end{bmatrix} \begin{bmatrix} \dot{c} \\ \dot{T} \\ \dot{u} \end{bmatrix} + \begin{bmatrix} K_{cc} & K_{cT} & K_{cu} \\ K_{Tc} & K_{TT} & K_{Tu} \\ K_{uc} & K_{uT} & K_{uu} \end{bmatrix} \begin{bmatrix} c \\ T \\ u \end{bmatrix} = \begin{bmatrix} f_c \\ f_T \\ f_u \end{bmatrix}$$

$$\begin{bmatrix} M_{cc} & 0 & 0 \\ 0 & M_{TT} & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} \dot{c} \\ \dot{T} \\ 0 \end{bmatrix} + \begin{bmatrix} K_{cc} & K_{cT} & K_{cu} \\ K_{Tc} & K_{TT} & K_{Tu} \\ K_{uc} & K_{uT} & K_{uu} \end{bmatrix} \begin{bmatrix} c \\ T \\ u \end{bmatrix} = \begin{bmatrix} f_c \\ f_T \\ f_u \end{bmatrix}$$

Consider all integral forms in weak form one-by-one:
 Consider the primal fields in terms of basis functions:
 elemental

$$c^e = \sum_{A=1}^{N_{ne}} N_c^A d_c^A$$

$$c^e = \sum_{A=1}^{N_{ne}} N_c^A d_c^A$$

$$T^e = \sum_{A=1}^{N_{ne}} N_T^A d_T^A$$

$$u_i^e = \sum_{A=1}^{N_{ne}} N_u^A d_{ui}^A$$

derivatives -

$$c_{,i}^e = \sum_{B=1}^{N_{ne}} N_{c,i}^B d_c^B$$

Shape Functions:

$$c^e = \sum_{A=1}^{N_{ne}} N_c^A d_c^A$$

$$T^e = \sum_{A=1}^{N_{ne}} N_T^A d_T^A$$

$$u_i^e = \sum_{A=1}^{N_{ne}} N_u^A d_{ui}^A$$

$$c_{,i}^e = \sum_{A=1}^{N_{ne}} N_{c,i}^A d_c^A$$

$$T_{,i}^e = \sum_{A=1}^{N_{ne}} N_{T,i}^A d_T^A$$

$$u_{i,j}^e = \sum_{A=1}^{N_{ne}} N_{u,i,j}^A d_{ui}^A$$

$N_{c,i}^A d_{c,i}^A$

comes from isoparametric mapping

~~W~~

$$w^e = \sum_{B=1}^{N_{ne}} N_c^B q_c^B$$

$$w_T^e = \sum_{B=1}^{N_{ne}} N_T^B q_T^B$$

$$w_i^e = \sum_{B=1}^{N_{ne}} N_u^B q_{ui}^B$$

$$w_{,i}^e = \sum_{B=1}^{N_{ne}} N_{c,i}^B q_c^B$$

$$w_{T,i}^e = \sum_{B=1}^{N_{ne}} N_{T,i}^B q_T^B$$

$$w_{i,j}^e = \sum_{B=1}^{N_{ne}} N_{u,i,j}^B q_{ui}^B$$

The three primal fields c , T , u can have diff. no. of nodes in an element, & hence ~~diff~~ can have different shape functions.

Isoparametric Mapping:

The primal fields c, T, u can have different basis fns & hence different isoparametric mappings.

OR

The geometry mapping can be fixed independently. It does not have to match any particular field.

We'll do that here —

Define geometry interpolation with its own functions $N^{geo}(\bar{x})$

$$x_i = \sum_{A=1}^{N_{ne}^{geo}} N_{i,j}^A x_i^A \quad x_i(\bar{x}) = \sum_{A=1}^{N_{ne}^{geo}} N_{i,j}^A(\bar{x}) x_i^A$$

given any point $\bar{x}_i$ in the parent domain, what is its equivalent in the physical domain?

The Jacobian of mapping —

$$\frac{\partial x_i}{\partial \bar{x}_j} = \sum_{A=1}^{N_{ne}^{geo}} N_{i,j}^A x_i^A$$

$$\text{Jacobian: } J = \begin{bmatrix} \frac{\partial x_1}{\partial \bar{x}_1} & \frac{\partial x_1}{\partial \bar{x}_2} \\ \frac{\partial x_2}{\partial \bar{x}_1} & \frac{\partial x_2}{\partial \bar{x}_2} \end{bmatrix}$$

$$J^{-1} = \begin{bmatrix} \frac{\partial \bar{x}_1}{\partial x_1} & \frac{\partial \bar{x}_1}{\partial x_2} \\ \frac{\partial \bar{x}_2}{\partial x_1} & \frac{\partial \bar{x}_2}{\partial x_2} \end{bmatrix}$$

That means $\rightarrow$ The Jacobian is calculated once per quadrature point using N_{geo} & A .

$\rightarrow$ Then J^{-1} is applied separately to the reference space derivative of N_c , N_T & N_u to get their physical-space derivative.

$\rightarrow$ Each field has its own shape functions in η -space, but they all share the same physical-space transformation.

Next: Consider all integral forms in weak form one-by-one:

Chemical:

$$\begin{aligned}
 \Rightarrow M_{cc} : \int_{\Omega} w^c \frac{dc}{dt} dV &= \sum_c \int_{\Omega} w^c \frac{dc^c}{dt} dV \\
 &= \sum_c \int_{\Omega} \left(\sum_{A=1}^{N_{ne}^c} N_c^A q_c^A \right) \cdot \left(\sum_{B=1}^{N_{ne}^c} N_c^B \dot{d}_c^B \right) dV \\
 &= \sum_c \sum_{A,B}^{N_{ne}^c} q_c^A \left(\int_{\Omega} N_c^A N_c^B dV \right) \dot{d}_c^B \\
 &= \sum_c \sum_{A,B}^{N_{ne}^c} q_c^A \left(\int_{\Omega^{\eta}} N_c^A(\eta) N_c^B(\eta) \det[J] dV^{\eta} \right) \dot{d}_c^B \\
 &\quad \underbrace{\hspace{10em}}_{M_{cc}^c (N_{ne}^c \times N_{ne}^c)}
 \end{aligned}$$

K_{cc}:

$$\int_{\Omega} \nabla w^c D_c (1 + F_{ijkl} \alpha_{ij}^c \alpha_{kl}^c) \nabla c \, dV = \sum_e \int_{\Omega_e} \nabla w^e D_c (1 + F_{ijkl} \alpha_{ij}^e \alpha_{kl}^e) \nabla c^e \, dV$$

$$\Rightarrow \sum_e \int_{\Omega_e} \left(\sum_{A=1}^{N_{ne}^c} N_{c,i}^A q_c^A \right) D_c (1 + F_{ijkl} \alpha_{ij}^e \alpha_{kl}^e) \cdot \left(\sum_{B=1}^{N_{ne}^c} N_{c,j}^B d_c^B \right) dV$$

$$\Rightarrow \sum_e \sum_{A,B}^{N_{ne}^c} q_c^A \left(\int_{\Omega_e} N_{c,i}^A D_c (1 + F_{ijkl} \alpha_{ij}^e \alpha_{kl}^e) N_{c,j}^B dV \right) \cdot d_c^B$$

$$\Rightarrow \sum_e \sum_{A,B}^{N_{ne}^c} q_c^A \underbrace{\left(\int_{\Omega_e} (N_{c,i}^A(z)) z_{,i} D_c (1 + F_{ijkl} \alpha_{ij}^e \alpha_{kl}^e) (N_{c,j}^B(z)) z_{,j} \det[J] dV^z \right)}_{K_{cc}^e (N_{ne}^c \times N_{ne}^c)} d_c^B$$

K_{CT}:

$$\int_{\Omega} \nabla w^c D_c \Phi_{ijkl} \alpha_{ij}^T \alpha_{kl}^c \nabla T^e dV = \sum_e \int_{\Omega_e} \nabla w^{ce} D_c \Phi_{ijkl} \alpha_{ij}^T \alpha_{kl}^c \nabla T^e dV$$

$$\Rightarrow \sum_e \int_{\Omega_e} \left(\sum_{A=1}^{N_{ne}^c} N_{c,i}^A q_c^A \right) D_c \Phi_{ijkl} \alpha_{ij}^T \alpha_{kl}^c \left(\sum_{B=1}^{N_{ne}^T} N_{T,j}^B d_T^B \right) dV$$

$$\Rightarrow \sum_e \sum_A^{N_{ne}^c} \sum_B^{N_{ne}^T} q_c^A \left(N_{c,i}^A D_c \Phi_{ijkl} \alpha_{ij}^T \alpha_{kl}^c N_{T,j}^B d_T^B \right) dV$$

$$\Rightarrow \sum_e \sum_A^{N_{ne}^c} \sum_B^{N_{ne}^T} q_c^A \left(\int_{\Omega_e} \left(N_{c,i}^A(\xi) \xi_{,i} \right) D_c \Phi_{ijkl} \alpha_{ij}^T \alpha_{kl}^c \left(N_{T,j}^B(\xi) \cdot \xi_{,j} \right) \det[J] dV^\xi \right) d_T^B$$

$K_{CT}^e \quad (N_{ne}^c \times N_{ne}^T)$

K_{cu}:

$$\int_{\Omega} w^c D_c F_{ijke} \alpha_{ke}^c \delta E_{ij} dV = \sum_c \int_{\Omega^c} w^c D_c F_{ijke} \alpha_{ke}^c$$

Thermal :

$$M_{II}: \int_{\Omega} w^T \frac{\partial T}{\partial t} dV = \sum_c \int_{\Omega^c} w^T \frac{\partial T^c}{\partial t} dV$$

$$= \sum_c \int_{\Omega^c} \left(\sum_{A=1}^{N_{ne}^T} N_T^A q_T^A \right) \cdot \left(\sum_{B=1}^{N_{ne}^T} N_T^B \dot{d}_T^B \right) dV$$

$$= \sum_c \sum_{A,B}^{N_{ne}^T} q_T^A \left(\int_{\Omega^c} N_T^A N_T^B dV \right) \dot{d}_T^B$$

$$= \sum_c \sum_{A,B}^{N_{ne}^T} q_T^A \left(\underbrace{\int_{\Omega^c} N_T^A N_T^B dV}_{M_{TT}^c (N_{ne}^T \times N_{ne}^T)} \right) \dot{d}_T^B$$

Q K_{TT} :

$$\int_{\Omega} \nabla w^T D_T (1 + \Phi_{ijkl} \alpha_{ij}^T \alpha_{kl}^T) \nabla T dV = \sum_e \int_{\Omega^e} \nabla w^{Te} D_T (1 + \Phi_{ijkl} \alpha_{ij}^T \alpha_{kl}^T) \nabla T^e dV$$

$$\Rightarrow \sum_e \int_{\Omega^e} \left(\sum_{A=1}^{N_{ne}^T} N_{T,i}^A q_T^A \right) D_T (1 + \Phi_{ijkl} \alpha_{ij}^T \alpha_{kl}^T) \left(\sum_{B=1}^{N_{ne}^T} N_{T,j}^B d_T^B \right) dV$$

$$\Rightarrow \sum_e \sum_{A,B}^{N_{ne}^T} q_T^A \left(\int_{\Omega^e} N_{T,i}^A D_T (1 + \Phi_{ijkl} \alpha_{ij}^T \alpha_{kl}^T) N_{T,j}^B dV \right) d_T^B$$

$$\Rightarrow \sum_e \sum_{A,B}^{N_{ne}^T} q_T^A \left(\int_{\Omega^e} \underbrace{\left(N_{T,i}^A(z) z_{,i} \right) D_T (1 + \Phi_{ijkl} \alpha_{ij}^T \alpha_{kl}^T) \left(N_{T,j}^B(z) z_{,j} \right) \det[J] dV^z}_{K_{TT}^e \quad (N_{ne}^T \times N_{ne}^T)} \right) d_T^B$$

K_{TC}:

$$\int_{\Omega} \nabla w^T D_T \Phi_{ijkl} \alpha_{ij}^c \alpha_{kl}^T \nabla c^e dV = \sum_e \int_{\Omega^e} \nabla w^{Te} D_T \Phi_{ijkl} \alpha_{ij}^c \alpha_{kl}^T \nabla c^e dV$$

$$\Rightarrow \sum_e \int_{\Omega^e} \left(\sum_{A=1}^{N_{ne}^T} N_{Ti}^A q_{LT}^A \right) D_T \Phi_{ijkl} \alpha_{ij}^c \alpha_{kl}^T \left(\sum_{B=1}^{N_{ne}^c} N_{cj}^B d_c^B \right) dV$$

$$\Rightarrow \sum_e \sum_A \sum_B q_{LT}^A \left(\int_{\Omega^e} N_{Ti}^A D_T \Phi_{ijkl} \alpha_{ij}^c \alpha_{kl}^T N_{cj}^B dV \right) d_c^B$$

$$\Rightarrow \sum_e \sum_A \sum_B q_{LT}^A \left(\int_{\Omega^e} \left(N_{Ti,i}^A \zeta_{i,i} \right) D_T \Phi_{ijkl} \alpha_{ij}^c \alpha_{kl}^T \left(N_{cj,j}^B \zeta_{j,j} \right) d\Omega [J] dV^J \right) d_c^B$$

$K_{TC}^e \quad (N_{ne}^T \times N_{ne}^c)$

Mechanics: let assume $E_{ke} = \frac{1}{2} \left(\frac{\partial u_k}{\partial x_e} + \frac{\partial u_e}{\partial x_k} \right) = \frac{\partial u_e}{\partial x_k} = u_{k,e}$

Kuu: $\int_{\Omega} w_{i,j}^u \phi_{ijkl} E_{ke} dV = \sum_e \int_{\Omega^e} w_{i,j}^{ue} \phi_{ijkl} u_{k,e}^e dV$

$$\Rightarrow \sum_e \int_{\Omega^e} \left(\sum_{A=1}^{N_{ne}^u} N_{u,j}^A q_{ui}^A \right) \phi_{ijkl} \left(\sum_{B=1}^{N_{ne}^u} N_{u,k}^B d_{u,k}^B \right) dV$$

$$\Rightarrow \sum_e \sum_{A,B}^{N_{ne}^u} q_{ui}^A \left(\int_{\Omega^e} N_{u,j}^A \phi_{ijkl} N_{u,k}^B dV \right) d_{u,k}^B$$

$$\Rightarrow \sum_e \sum_{A,B}^{N_{ne}^u} q_{ui}^A \left(\int_{\Omega^e} (N_{u,j}^A \zeta_{i,j}) \phi_{ijkl} (N_{u,k}^B \zeta_{k,l}) \det[J] dV^{\zeta} \right) d_{u,k}^B$$

K_{uu}^e $(N_{ne}^u \times N_{ne}^u)$
 $N_{nd}^u \times N_{ne}^u$

Kuc:

$$-\int_{\Omega} w_{ij}^u \phi_{ijke} \alpha_{ke}^c dV = - \sum_e \int_{\Omega^e} w_{ij}^{ue} \phi_{ijke} \alpha_{ke}^c dV$$

$$\Rightarrow - \sum_e \int_{\Omega^e} \left(\sum_{A=1}^{N_{ne}^u} N_{u,j}^A q_{ui}^A \right) \phi_{ijke} \alpha_{ke}^c \left(\sum_{B=1}^{N_{ne}^c} N_c^B d_c^B \right) dV$$

$$\Rightarrow - \sum_e \sum_A^{N_{ne}^u} \sum_B^{N_{ne}^c} q_{ui}^A \left(\int_{\Omega^e} N_{u,j}^A \phi_{ijke} \alpha_{ke}^c N_c^B dV \right) d_c^B$$

$$\Rightarrow - \sum_e \sum_A^{N_{ne}^u} \sum_B^{N_{ne}^c} q_{ui}^A \left(- \int_{\Omega^e} \left(N_{u,j}^A(\xi) \xi_{i,j} \right) \phi_{ijke} \alpha_{ke}^c \left(N_c^B(\xi) \right) \det[J] dV^{\xi} \right) d_c^B$$

$$K_{uc}^e \quad (N_{ne}^u \times N_{ne}^c)$$

$\downarrow$
 $N_{nd}^u \times N_{ne}^u$

Aux: $\Lambda_{ij}^c = \phi_{ijke} \alpha_{ke}^c = \lambda \text{tr}(\alpha^c) \mathbb{I} + 2\mu \alpha^c$

$\gamma_{voigt}^c = \epsilon_{voigt}^c \cdot \alpha_{voigt}^c$

$\downarrow$ (6x6) $\downarrow$ (6x1)
 $\downarrow$

$$\alpha_{voigt}^c = [\alpha_{11}^c, \alpha_{22}^c, \alpha_{33}^c, \alpha_{12}^c, \alpha_{13}^c, \alpha_{23}^c]^T$$

K_{UT}:

$$-\int_{\Omega} \omega_{ij}^u \phi_{ijk} \alpha_{kl}^T T dV = -\sum_e \int_{\Omega^e} \omega_{ij}^{ue} \phi_{ijk} \alpha_{kl}^T T^e dV$$

$$\Rightarrow -\sum_e \int_{\Omega^e} \left(\sum_A^{N_{ne}^u} N_{u,j}^A q_{ui}^A \right) \phi_{ijk} \alpha_{kl}^T \left(\sum_B^{N_{ne}^T} N_T^B d_T^B \right) dV$$

$$\Rightarrow -\sum_e \sum_A^{N_{ne}^u} \sum_B^{N_{ne}^T} q_{ui}^A \left(\int_{\Omega^e} N_{u,j}^A \phi_{ijk} \alpha_{kl}^T N_T^B dV \right) d_T^B$$

$$\Rightarrow -\sum_e \sum_A^{N_{ne}^u} \sum_B^{N_{ne}^T} q_{ui}^A \left(- \int_{\Omega^e} (N_{u,j}^A(z)) \phi_{ijk} \alpha_{kl}^T (N_T^B(z)) \det[J] dV^z \right) d_T^B$$

K_{UT}^e ($N_{ne}^u \times N_{ne}^T$)

N_{sd}^u of N_{ne}^u

Aux:

$$\Lambda_{ij}^T = \phi_{ijk} \alpha_{kl}^T = \lambda \text{tr}(\alpha^T) \mathbb{I} + 2\mu \alpha^T$$

Flow Chart

① Initialize $c^0, T^0, u^0 = 0$

② Build mesh, BCs, quadrature, material

③ Time loop: $0 \rightarrow NT$

④ Diffusion Assembler : evaluate $M_{cc}, M_{TT}, K_{cc}, K_{TT}, K_{cT}, K_{Tc}, f_c, f_T$
 $\uparrow$
 come from Element Evaluator.

⑤ Compute c & T residuals, & update them

Residuals:

$$R^{c,n} = f^{c,n} - K_{cc}c^n - K_{cT}T^n$$

$$R^{T,n} = f^{T,n} - K_{Tc}c^n - K_{TT}T^n$$

Update:

$$c^{n+1} = c^n + \Delta t \cdot M_{cc}^{-1} R^{c,n}$$

$$T^{n+1} = T^n + \Delta t M_{TT}^{-1} R^{T,n}$$

⑥ Freeze $c^{n+1}, T^{n+1} \rightarrow$ pass to mechanics (Newton-Raphson Solver for u)

for each iter k :

Goal: find $P \rightarrow K$

$\swarrow$ SVK Material Model
 Assemble into $R_{global}^u, K_{global}^u$

$\downarrow$
~~Partition~~ Add eigenstrain loads (effect of c on u , T on u)
 $R_{global}^u = R_{global}^u - K_{uc}c^{n+1} - K_{uT}T^{n+1}$

$\downarrow$
 Partition: R_U, K_{UU}^u

$\downarrow$
 Solve $K_{UU} \cdot \Delta u = -R_U$ (sparse LU/CG)

$\downarrow$
 $u^{k+1} = u^k + \Delta u$

$\downarrow$
 check $\|R_U^u\| < \epsilon \rightarrow$ converged

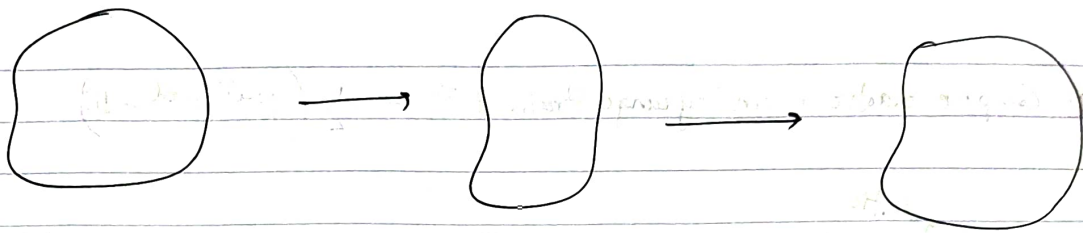
write off $c^{n+1} \rightarrow c^n$

$T^{n+1} \rightarrow T^n$

$u^{n+1} \rightarrow u^n$

$\downarrow$
 Repeat from ④

Finite Strain Material Model



Reference Ω_0

undeformed body
no loads, no swelling,
no temp. change.

Coordinates: x

Intermediate Ω'

body after chemical
swelling (F^{ch}) &
thermal expansion (F^{th})

but
before elastic deformⁿ

Current Ω

actual deformed body
→ reached from Ω' by
elastic deformⁿ F^{el}

Coordinates: $x = \phi(x)$

total deformⁿ gradient: $F = \frac{d\phi(x)}{dx}$

$$F = F^{el} \cdot F^{th} \cdot F^{ch}$$

Note: the material only stores elastic energy. The chemical & thermal parts expand the body, but produce no stress on their own. They just change the stress-free reference for the elastic part.

what Material Model does: AT EACH QUADRATURE POINT:

① Compute F^{th} & F^{ch} from T^{n+1} & C^{n+1}

volumetric thermal ratio J^{th}

$$J^{th} = 1 + \alpha T^{n+1}$$

stretch

$$F^{th} = (J^{th})^{1/3}$$

stretch

$$F^{th} = I + \alpha T$$

volumetric swelling ratio J^{ch}

$$J^{ch} = 1 + \alpha^c C^{n+1}$$

stretch

$$F^{ch} = (J^{ch})^{1/3}$$

stretch

$$F^{ch} = I + \alpha^c C$$

② Extract F^{el} from total F using F^{th} & F^{ch}

$$F^{el} = F \cdot F^{th-1} \cdot F^{ch-1}$$

$$F^{el} = \frac{F}{(J^{th})^{1/3} (J^{ch})^{1/3}}$$

③ Compute elastic Green Lagrange strain $E^{el} = \frac{1}{2} (F^{elT} \cdot F^{el} - \mathbb{I})$

④ Compute ^{elastic} Second Piola Kirchhoff stress $S_{ij}^{el} = K_{ijkl} E_{kl}^{el} = 2\mu E^{el} + \lambda \text{tr}(E) \mathbb{I}$

S^{el} is defined in intermediate configuration. ~~To get~~

To get second PK stress, in reference configⁿ, we pull it back thru F^{ch} & F^{th}

$$S = (F^{ch})^{-1} (F^{th})^{-1} S^{el} (F^{th})^T (F^{ch})^T$$

⑤ Compute first PK stress $P = F \cdot S$